

Reduction of a Six-Coordinate Representation to a Four-Coordinate Space Using Barycentric Coordinates and 4x4 Determinants

Roman Kotlowski

September 18, 2026

Abstract

This work presents a numerical method for reducing a six-coordinate description, based on three pairs of parameters (AL_i, AN_i) , to a representation in a four-coordinate space. The method uses barycentric coordinates and determinants of 4×4 matrices. The implementation is available at: <https://github.com/rangeman1/GCR-4>.

1 Introduction

In many computational problems, there is a need to reduce the complexity of a multi-coordinate description to a more compact form while preserving essential geometric relationships between the data. In the presented method, the starting point is a six-coordinate state representation, expressed through three parameter pairs (AL_i, AN_i) , which may be interpreted as coordinates along two independent axes for three measurement points or three system features.

The goal of the method is to transform this description into a four-coordinate representation, where each point is described by coefficients (R_1, R_2, R_3, R_4) . These coefficients act as barycentric coordinates in a four-coordinate rectangular simplex and provide a compact and numerically stable representation of the input data.

The reduction process consists of two stages. In the first step, each pair (AL_i, AN_i) is mapped to a triple (X_i, Y_i, Z_i) satisfying the barycentric condition

$$X_i + Y_i + Z_i = 1.$$

This transformation corresponds to projecting the point onto a two-dimensional simplex (triangle), providing a local flattening of the input data.

In the second step, the three obtained triples (X_i, Y_i, Z_i) are used to construct the main matrix M_{main} and four auxiliary matrices in which selected columns are replaced with barycentric-related values. The determinants of these matrices — W and $W_{X1}, W_{X2}, W_{X3}, W_{X4}$ — allow computing the coefficients

$$R_k = \frac{W_{Xk}}{W}, \quad k = 1, 2, 3, 4.$$

These coefficients serve as coordinates in the four-coordinate space and constitute the final result of the reduction. The method is deterministic, based on classical linear algebra tools, and does not require iterative optimization procedures.

2 Method

2.1 Barycentric Coordinates

Each pair (AL, AN) is transformed into a triple (X, Y, Z) satisfying:

$$X + Y + Z = 1.$$

2.2 Matrices and Determinants

The main matrix has the form:

$$M_{\text{main}} = \begin{pmatrix} 1 - X_1 & -X_1 & -X_1 & 0 \\ 0 & 1 - X_2 & -X_2 & -X_2 \\ 1 - X_3 & 0 & -X_3 & -X_3 \\ 1 & 1 & 1 & 1 \end{pmatrix}.$$

The reduction coefficients are computed as:

$$R_k = \frac{W_{Xk}}{W}.$$

3 Summary

The presented method enables deterministic reduction from a six-coordinate space described by parameter pairs (AL_i, AN_i) to a four-coordinate representation (R_1, R_2, R_3, R_4) .